INCH-POUND

MIL-DTL-45225G (MR)
25 April 2025
SUPERSEDING
MIL-DTL-45225F (MR)
22 April 1998

DETAIL SPECIFICATION

ALUMINUM ALLOY ARMOR, FORGED

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers forged aluminum alloy armor, die forgings from 3/4 inch up to and including 3 inches in thickness for type I, class A forgings, and from 3/4 inch up to and including 4 inches for type I, class B forgings. Type II forging thicknesses should be as specified (see 6.2). For qualification or lot release verification on forgings with thicknesses greater than the current limits, testing is allowed on machined parts or milled samples at a thinner gauge. The tested thickness and the performance threshold will be determined by the purchasing agent at time of contract or purchase order.

1.2 Classification Forgings furnished under this specification are furnished in one of the following types and classes.

1.2.1 Type I. Forgings manufactured using weldable alloys.

1.2.1.1 Class A. Type I, class A includes weldable non-heat treatable alloys 5083 and 5456.

1.2.1.2 Class B. Type I, class B includes weldable heat treatable alloys 7039, Al-Zn-Mg, and Al-Mg-Zn alloys.

1.2.2 Type II. Forgings manufactured using non-weldable alloys.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A AREA FORG

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662	-	V_{50} Ballistic Test for Armor
MIL-STD-3057	-	Arc Welding of Armor Grade Aluminum

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE ALUMINUM ASSOCIATION, INC.

Teal Sheets	-	International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys

(Copies of these documents are available online at https://www.aluminum.org/).

ASTM INTERNATIONAL

ASTM B557	-	Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products
ASTM E716	-	Standard Practices for Sampling and Sample Preparation of Aluminum and Aluminum Alloys for Determination of Chemical Composition by Spark Atomic Emission Spectrometry
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Spark Atomic Emission Spectrometry

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ASTM E3061	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)

(Copies of these documents are available online at https://www.astm.org/.)

SAE INTERNATIONAL

AMS 2750	-	Pyrometry
AMS 2772	-	Heat Treatment of Aluminum Alloy Raw Materials

(Copies of these documents are available online at https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required, and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

3.2 Chemical composition. The chemical compositions for type I alloys defined by the Aluminum Association Teal Sheets, “International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys” are listed in table I and shall be within those limits. A certification of conformance of the chemical composition of the alloy used in the forging shall be furnished. Type II material shall be as specified in the contract or on the drawings.

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TABLE I. Type I alloy chemical composition, percent. 1/

ELEMENTS	SYMBOL	Class A 5083 2/	Class A 5456 2/	Class B 7039 2/	Class B Al-Zn-Mg 2/
Silicon	Si	0.40	0.25	0.45	0.35
Iron	Fe	0.40	0.40	0.50	0.45
Copper	Cu	0.10	0.10	0.25	0.20
Manganese	Mn	0.40 - 1.0	0.50 - 1.0	0.6 - 1.2	0.05 - 0.70
Magnesium	Mg	4.0 - 4.9	4.7 - 5.5	5.0 - 6.0	1.0 - 3.8
Chromium	Cr	0.05 - 0.25	0.05 - 0.20	0.25	0.10 - 0.35
Zinc	Zn	0.25	0.25	0.40 - 0.9	4.0 - 5.2
Titanium	Ti	0.15	0.20	0.20	0.15
Zirconium	Zr	---	---	0.05 - 0.25	0.08 - 0.25
Other, max. Each	---	0.05	0.05	0.05	0.05
Other, max. Total 3/	---	0.15	0.15	0.15	0.15
Aluminum	Al	Remainder	Remainder	Remainder	Remainder

1/ Except for "Aluminum" and "others", analysis normally is made for elements for which specific limits are shown.
2/ Where single units are shown, these indicate the maximum amounts permitted.
3/ The sum of those "others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

3.3 Mechanical properties. Unless otherwise specified (see 6.2), the mechanical properties of type I forgings shall meet the requirements of table II, when tests are conducted as specified in 4.8.2. Type II forgings shall meet the mechanical property requirements specified in the contract or on the drawing. If mechanical properties different from the values shown in table II or any other properties are required, the ballistic requirements must then be negotiated (see 3.4).

TABLE IIa. Minimum longitudinal mechanical properties. 1/,2/

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Class A	Class B	Class A	Class B	Class A	Class B
0.750 to 3.000, incl.	42.0	57.0	33.0	48.0	7	8

TABLE IIa. Minimum transverse mechanical properties. 1/,2/

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Class A	Class B	Class A	Class B	Class A	Class B
0.750 to 3.000, incl.	40.0	54.0	32.0	45.0	4	4

3.4 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix A. No ballistic limit shall be specified for forging thickness of 0.250 to 0.749 inch, inclusive. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the

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specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.5 Thermal processing, type I.

3.5.1 Class A. After final strain hardening performed during the forging process, any thermal processing exceeding 200 °F (93.3 °C) performed on the forging, excluding fabrication by welding in accordance with MIL-DTL-3057, will necessitate re-verification for conformance with 3.3 and 3.4. Deviation from this requirement shall be subject to review and approval by the procuring activity.

3.5.2 Class B. Heat treatment shall conform to the requirements of SAE AMS 2750 and SAE AMS 2772 and shall be such as to enable the material to meet the requirements of these specifications.

3.6 Dimensional tolerances. The forgings shall conform to the shape and dimensions specified on the drawings within any variations stated herein.

3.7 Marking for identification. Each forging shall be marked in at least one location in characters not less than 3/8 inch high with the following information. Impression stamping shall not be used, except in areas designated by the drawing.

(a) Manufacturer's name, trademark or both.
(b) Number of this specification.
(c) Type, class, alloy designation, and temper.
(d) Army part or drawing number.
(e) Lot number.

3.7.1 Ballistic Test Forgings. First article and acceptance and ballistic test forgings shall be marked with the information in 3.7. In addition, first article forgings shall be marked "PRE" and acceptance test forgings marked "ACC". This marking shall be impression stamped in at least two opposing locations in letters not less than 1/2-inch high.

3.8 Information required on ballistic test forgings. Unless otherwise specified in the contract or purchase order, a properly completed Aluminum Alloy Armor, Forged Test Data Form (see figure 1) shall be submitted with each ballistic test plate or forging containing the information as specified in MIL-STD-662. Ballistic retest forgings shall be marked "R1" and "R2" respectively (see A.5.2).

3.9 Workmanship. The forgings shall be uniform in quality and free from blisters, fins, seams, laps, and cracks.

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REQUEST FOR BALLISTIC TEST OF ALUMINUM ARMOR FORGING										
SPECIFICATION: MIL-DTL-45225			**REVISION: G**			**AMENDMENT:**				
FIRING RECORD:		**DATE:**			**TEST CONTRACT NO:**					
	FORGING PRODUCER	**HEAT TREATER**		**PRIME CONTRACTOR**						
Name:										
Address:										
POC:										
Phone No:										
Email:										
TEST ITEM IDENTIFICATION:				**Ordered Thickness:**						
Lot No. _____ **Forging No.** _____		**Alloy:** _____ **Temper:** _____			**Type:** _____ **Class:** _____					
PURPOSE: ___ Acceptance ___ First Article ___ Development ___ Reduced Testing ___ Audit Testing										
SAMPLE: ___ Primary ___ Retest (Firing Record No. of Failed Sample _____)										
CHEMISTRY ANALYSIS:	**Si**	**Fe**	**Cu**	**Mn**	**Mg**	**Cr**	**Zn**	**Ti**	**Zr**	
V	**Li**	**Ag**	**Ni**	**Mn+Cr**	**Zr+Ti**	**Other - Each**			**Others - Total**	**Al**
										Rem.
MECHANICAL PROPERTIES:										
Longitudinal	UTS (ksi):		0.2% YS (ksi):		Elongation (%):					
Transverse	UTS (ksi):		0.2% YS (ksi):		Elongation (%):					
BALLISTIC TEST RESULTS: SEPARATELY HEAT-TREATED TEST FORGING : YES ____ NO ____										
	Obl. (deg)	**Actual Thickness (in)**	**Required V_{50} (fps)**	**Actual V_{50} (fps)**	**Pass/ Fail**					
Test:										
Projectile:										
	NOTES:									
SPALL:	Spall Size:		Remarks:							
DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY GOVERNMENT & SUPPLIER OFFICIALS										
LOT: [____ met] [____ failed to meet] the ballistic requirements of specification MIL-DTL-45225										
Government Representative:		Date:	Supplier Representative:		Date:					

FIGURE 1. Aluminum Alloy Armor, Forged Test Data Form.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Lot. A lot shall consist of forgings of the same shape, alloy, and processing submitted for inspection as a unit, provided that the lot does not exceed 25,000 pounds or 300 forgings, whichever is less.

4.3 First article. First article inspection, except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4. First article inspection shall consist of the following:

(a) Chemical analysis (see 3.2 and 4.7.3).
(b) Mechanical properties (see 3.3 and 4.8.2).
(c) Ballistic tests (see 3.4 and 4.8.3).
(d) Dimensions (see 3.6 and 4.6.2).
(e) Identification marking (see 3.7 and 4.6.1).
(f) Workmanship (see 3.9 and 4.6.1).

4.4 Conformance (acceptance) inspection. Test for acceptance of individual lots consist of testing all requirements specified in section 3. Verification conformance inspection is in accordance with the provisions herein. Lot acceptance inspection shall consist of the following:

(a) Chemical analysis (see 3.2 and 4.8.1).
(b) Mechanical properties (see 3.3 and 4.8.2).
(c) Ballistic tests (see 3.4 and 4.8.3).
(d) Dimensions (see 3.6 and 4.6.2).
(e) Identification marking (see 3.7, 3.7.1, and 4.6.1).
(f) Workmanship (see 3.9 and 4.6.1).
(g) Preparation for shipment (see 4.6.3 and section 5).

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examination of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1 First article inspection. At least three sample forgings shall be submitted for ballistic and mechanical testing. If the forging size is too small for removal of tensile test specimens from the ballistic sample, three additional forgings shall be furnished.

4.5.1.1 Chemical composition. For type I forgings, a certification of compliance with the limits shown in table I shall be furnished with the test forgings. For type II forgings, a certification of

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compliance with the chemical composition limits stated in the contract shall be furnished with the test forgings.

4.5.1.2 Mechanical properties. One longitudinal and one transverse tension test specimen shall be removed from each forging. If size permits, this may be the same forging that is used for ballistic tests. Location and type of specimens shall be as specified in 4.7.4.

4.5.1.3 Ballistic tests. For each configuration that the manufacturer has under contract, three or more forgings shall be submitted for ballistic testing in accordance with Appendix A. Prior to submission of the forgings, the tester and the contractor will discuss, using a drawing of the forgings, the ballistic test conditions; namely, forging thickness, impact area and projectile type. The number of forgings, or forging sections needed for first article ballistic acceptance testing will be determined. Modifications, cutting, welding and bracketing will be determined as needed for test mounting.

4.5.2 Acceptance tests.

4.5.2.1 Chemical composition, ingot analysis. At least one sample shall be taken from the molten metal representing one group of ingots poured as a unit from the same source of molten metal. Complete ingot analysis records shall be available to the government at the ingot producer's facility.

4.5.2.2 Chemical composition, finished product analysis. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.3 Mechanical properties. One forging for tension test shall be selected from each lot. One longitudinal and one transverse tension test specimen shall be removed from the forging. If size permits, this may be the same forging that is used for ballistic tests.

4.5.2.4 Ballistic testing. From each lot, one forging shall be submitted for ballistic test. (See paragraph 4.5.1.3 for supplementary information).

4.6 Examination.

4.6.1 Visual. Each forging shall be carefully examined to determine compliance with the identification marking (see 3.7) and workmanship (see 3.9) requirements.

4.6.1 Dimensional. For each acceptance lot, representative forgings shall be selected in accordance with table III and examined to determine compliance with requirements of 3.6.

TABLE III. Sampling for dimensional examination.

Lot size, pounds	Minimum number of samples 1/
Up to 50, incl.	5
51 t0 130, incl.	10
131 to 300, incl.	15

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4.6.2 Preparation for shipment. Prior to shipment, examination shall be made to determine compliance with section 5.

4.7 Test specimens.

4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E716, E1251, and E3061. In case of any dispute, analysis by method E3061 shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall conform to the dimensions of the standard 0.500 inch round specimen (figure 8) of ASTM B557. If the forging dimensions prevent the use of the standard 0.500 inch round specimen, the largest of the subsize proportional specimens shown in figure 8 of ASTM B557, which can be obtained, shall be used. Longitudinal tension test specimens shall be taken with the long axis parallel to the direction of maximum metal flow. Transverse specimens shall be taken with the long axis perpendicular to the direction of maximum metal flow. Actual location shall be as specified on the drawing or in the contract or order (see 6.2).

4.7.3 Ballistic testing.

4.7.3.1 Forging thickness measurement. When the forging thickness aligns within the range defined for ballistic requirements (see 1.1), the forging thickness used by the ballistic test agency to determine the required minimum protection ballistic limit shall be the measured thickness of the forging unless otherwise specified in the contract or purchase order. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 inch and rounded off to the nearest 0.005 of an inch. These measurements shall be made in regions of uniform thickness as close as possible to the intended point of impact of projectiles within the area required to provide ballistic protection as shown on the drawing. Forging thickness, as measured by the ballistic testing agency, shall be used in conjunction with table IV and Appendix A to determine the required V_{50} protection ballistic limit for that forging.

TABLE IV. Ballistic test forgings.

Nominal design thickness specified on drawing, (inches)		Projectile	Table
Class A	Class B		
0.705-1.100	0.705 - 1.030	Cal. .50 FSP	A-I
1.101-1.700	1.031 - 1.500	Cal. .30 AP	A-II
1.101-1.700	1.031 - 1.500	20-mm FSP	A-III
1.701-3.000	1.501 - 3.000	Cal. .50 AP M2	A-IV
---	3.001 - 4.000 1/	14.5-mm BS41	A-V

1/ 14.5 mm BS41 BL(P) data is unknown for this thickness range and to be determined (TBD).

4.7.3.2 Ballistic tests. Ballistic test forgings shall be tested at locations specified on the drawings or in the contract or order (see 6.2) at the government testing activity designated by the procuring activity. The forging thickness used by the ballistic test agency to determine the required minimum protection ballistic limit shall be the measured thickness (see 4.7.3.1) of the forging. The test

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forgings may be machined at the location specified for testing provided that such machining shall not cause the forgings to fall outside the thickness limits for the thickness range as specified in table IV. For forgings with thicknesses greater than the current limits, testing is allowed on machined parts or milled samples to attain a thinner gauge suitable for testing. The tested thickness and the performance threshold shall be determined by the purchasing agent at time of contract or purchase order. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an interpolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL (P) for that forging.

4.7.3.3 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059.

4.7.3.4 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

4.8 Rejection and retest. Unless otherwise specified in the contract or purchase order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, 4.8.2, and 4.8.3.

4.8.1 Rejection of first article forgings. Failure of the first article test forgings to meet the requirements of the examinations and tests listed in 4.3.1 indicates failure of the product. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test forging or specimens shall be subject to rejection, except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. Forgings submitted for retest shall be marked R1, R2 and R3 in addition to the requirements of 3.7. If one of the retest specimens fails, the lot shall be permanently rejected with no further testing permitted.

4.8.2 Retests.

4.8.2.1 Chemical composition and mechanical properties. Retests shall be conducted in accordance with one or more ASTM methods of E716, E1251, and E3061.

4.8.2.5 Ballistic rejection and retest. Rejection and retest of ballistic test forgings shall be in accordance with A.5.2.

4.9 Reduced testing. At the discretion of the procuring activity, the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced, and providing the manufacturer agrees in writing to maintain the

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same manufacturing procedures. The manufacturer when eligible for reduced or audit testing may at his option, continue to submit one ballistic test plate or forging per lot of material. The test facility shall only test the non-required test plates or forgings upon manufacturer's loss of reduced testing eligibility.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The aluminum alloy armor forgings covered by this specification are military unique. The forgings are used in military vehicles to protect the operator and passengers against small arms fire and shrapnel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Specify type, class and alloy (see 1.2).
(c) When first article is required (see 3.1 and 4.3).
(d) Mechanical properties, thickness, and chemical composition of type II forgings (see 1.1, 3.2, and 3.3).
(e) Mechanical property requirements of type I forgings if different from table II (see 3.3).
(f) Drawings, Dimensions, and tolerance requirements (see 3.6).
(g) If markings are different and when impression stamping is permissible (see 3.7, and 3.7.1).
(h) Define surface cracks, edge cracks and edge laminations for the specific application (see 3.11.1).
(i) If the weight of finished plate can exceed 25,000 pounds or 300 forgings (see 4.2).
(j) Dimensional sampling procedure approved by the procuring activity (see 4.6.1).
(k) If approval was requested and received for a different ballistic testing facility (see 4.7.3.3).
(l) Rejection and retest requirements, if different (see 4.8).
(m) If reduced testing is allowed (see 4.9).

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(n) Packaging requirements (see 5.1).

6.3 Ownership of ballistic test forgings.

6.3.1 First article and acceptance test forgings. First article and acceptance test forgings that comply with the requirements of this specification are considered as part of the lot of forgings that they represent, and ownership of them passes to the government with acceptance of that lot. First article and acceptance test forgings that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the manufacturer as does the rejectable lot which they represent.

6.3.1.1 Failure of forgings. At the time of shipment of the first article and acceptance test forgings, the government test activity should be informed by the manufacturer, in writing, as to the disposition of first article and acceptance test forgings which fail the test. If instructions for disposal of the tested forgings are not received after 15 days, the testing activity will exercise the prerogative of disposing of the material.

6.4 Density. The density of the specified alloys listed in this specification are as follows:

AA5083 = 0.096 lb/in^3 (2.66 g/cc)
AA5456 = 0.096 lb/in^3 (2.66 g/cc)
AA7039 = 0.099 lb/in^3 (2.74 g/cc)

6.5 Mechanical properties to ballistic requirements. The minimum mechanical properties specified (see 3.3) may not assure that aluminum armor forgings meet the specified ballistic requirements (see 3.4 and Appendix A).

6.6 Stress Corrosion. To prevent failure in service due to stress corrosion, care should be exercised in design and fabrication to minimize sustained surface tensile stresses in the transverse direction. Any machining performed on the forgings supplied according to this specification should be closely controlled and the procuring activity should specify protective measures, when necessary, such as shot peening, buttering, etc., to curtail any possibility of stress-corrosion cracking. No alkaline etching should be permitted.

6.7 Long Term Storage. Type I, class A (5XXX) Al-Mg alloys are susceptible to degradation of mechanical properties and corrosion resistance after extended durations at elevated temperatures beyond 131°F (55°C) and additional preparations for long term storage are recommended. Due to surface heating produced by direct sunlight and/or cumulative trapped heat in poorly ventilated CONEX type shipping and storage containers, ventilated indoor warehouse storage is recommended for optimal storage of MIL-DTL-44225 type I, class A qualified forgings. Outdoor storage that provides overhead shading from direct sunlight is also acceptable. Large, stacked forgings should be separated by nonmetallic spacers to facilitate easy access and manipulation by heavy equipment and to provide optimum airflow to dissipate heat and maintain lower temperatures.

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6.8 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

6.9 Subject term (key word) listing.

Ballistic limit
Ballistic testing
Caliber .30 AP M2
Caliber .50 AP M2
Caliber .50 FSP
Class A material
14.5-mm API
Class B material
M1114 HMMWV
Military vehicles
Stress corrosion
20-mm FSP
Type II material

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

ALUMINUM ALLOY ARMOR, FORGED

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of forged aluminum alloy armor, die forgings from 3/4 inch up to and including 3 inches in thickness for type I, class A forgings, and from 3/4 inch up to and including 4 inches for type I, class B forgings type II forging thicknesses as specified (see 6.2) when tested in accordance with the provisions of this specification. For qualification or lot release verification on forgings with thicknesses greater than the current limits, ballistic testing is allowed on machined parts or milled samples at a thinner gauge. The tested thickness and the ballistic limit will be determined by the purchasing agent at time of contract or purchase order. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification, and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of this document are available online at https://quicksearch.dla.mil/).

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

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APPENDIX A

A.3.4 Partial penetration, (PP). Any impact which is not a complete penetration may be considered a partial penetration.

A.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables A-I through A-V.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which

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is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of any of the first article test forgings to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest forgings shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest forgings shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test forging to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test forging representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at their own expense two additional test forgings from the same lot for ballistic retest. If either of these forgings fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test forgings. If either of these plates fails, the lot shall be permanently rejected.

A.5.3 Disposal of ballistic test forgings.

A.5.3.1 First article test forgings. Upon request of the applicant within 15 days after ballistic testing, first article forgings shall be returned "as is" to the applicant, at their own expense, unless the forgings were destroyed in testing.

A.5.3.2 Acceptance test forgings. Acceptance test forgings that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test forgings passes to the Government with the acceptance of that lot. Acceptance test forgings that fail to comply with the requirements of this specification are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed forgings shall be returned, upon request, as in A.5.3.1.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
0.650	1290	---	0.780	1760	1849
0.655	1308	---	0.785	1780	1870
0.660	1325	---	0.790	1800	1890
0.665	1343	---	0.795	1818	1911
0.670	1360	---	0.80	1835	1932
0.675	1378	---	0.805	1855	1953
0.680	1395	---	0.810	1875	1975
0.685	1413	---	0.815	1893	1996
0.690	1430	---	0.820	1910	2018
0.695	1448	---	0.825	1930	2040
0.700	1465	1550	0.830	1950	2063
0.705	1483	1567	0.835	1968	2085
0.710	1500	1584	0.840	1985	2108
0.715	1520	1602	0.845	2005	2131
0.720	1540	1620	0.850	2025	2154
0.725	1558	1638	0.855	2045	2178
0.730	1575	1656	0.860	2065	2201
0.735	1595	1675	0.865	2085	2225
0.740	1615	1693	0.870	2105	2249
0.745	1633	1712	0.875	2123	2274
0.750 1/	1650	1731	0.880	2140	2299
0.755	1668	1750	0.885	2160	2324
0.760	1685	1770	0.890	2180	2349
0.765	1705	1789	0.895	2200	2374
0.770	1725	1809	0.90	2220	2400
0.775	1743	1829	0.905	2240	2426

1/ Specification type I, class A and type I, class B requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
0.910	2260	2452	1.035	2775	3202
0.915	2280	2479	1.040	2795	3236
0.920	2300	2505	1.045	2818	3270
0.925	2320	2532	1.050	2840	3305
0.930	2340	2560	1.055	2863	3340
0.935	2360	2587	1.060	2885	3375
0.940	2380	2615	1.065	2908	3411
0.945	2400	2643	1.070	2930	3446
0.950	2420	2672	1.075	2950	3482
0.955	2440	2700	1.080	2970	3519
0.960	2460	2730	1.085	2993	3555
0.965	2480	2759	1.090	3015	3593
0.970	2500	2788	1.095	3038	3630
0.975	2523	2818	1.100 3/	3060	3668
0.980	2545	2849	1.105	3080	---
0.985	2565	2879	1.110	3100	---
0.990	2585	2910	1.115	3120	---
0.995	2605	2941	1.120	3140	---
1.000	2625	2972	1.125	3160	---
1.005	2648	3004	1.130	3180	---
1.010	2670	3036	1.135	3200	---
1.015	2690	3069	1.140	3220	---
1.020	2710	3102	1.145	3240	---
1.025	2733	3135	1.150	3260	---
1.030 2/	2755	3168	1.155	---	---

2/ Specification type I, class B requirements end for this ordered thickness.
3/ Specification type I, class A requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits - caliber 20 mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
1.000	1260	1401	1.150	1568	1649
1.005	1271	1409	1.155	1579	1658
1.010	1281	1416	1.160	1589	1666
1.015	1291	1424	1.165	1599	1675
1.020	1301	1432	1.170	1609	1685
1.025	1312	1440	1.175	1620	1694
1.030	1322	1447	1.180	1630	1703
1.031 [1/]	1324	1449	1.185	1640	1712
1.035	1332	1455	1.190	1650	1721
1.040	1342	1463	1.195	1661	1731
1.045	1353	1471	1.200	1671	1740
1.050	1363	1479	1.205	1681	1749
1.055	1373	1487	1.210	1692	1758
1.060	1384	1495	1.215	1702	1768
1.065	1394	1504	1.220	1712	1778
1.070	1404	1512	1.225	1722	1787
1.075	1414	1520	1.230	1733	1797
1.080	1425	1528	1.235	1743	1807
1.085	1435	1537	1.240	1753	1816
1.090	1445	1545	1.245	1763	1826
1.095	1455	1553	1.250	1774	1836
1.100	1466	1562	1.255	1784	1846
1.101 [2/]	1468	1564	1.260	1794	1855
1.105	1476	1570	1.265	1804	1865
1.110	1486	1579	1.270	1815	1875
1.115	1496	1587	1.275	1825	1885
1.120	1507	1596	1.280	1835	1896
1.125	1517	1605	1.285	1846	1906
1.130	1527	1613	1.290	1856	1916
1.135	1538	1622	1.295	1866	1926
1.140	1548	1631	1.300	1876	1937
1.145	1558	1640	1.305	1887	1947

[1/] Specification type I, class B requirements begin for this ordered thickness.
[2/] Specification type I, class A requirements begin at ordered thickness of 1.101 inches.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-II. Minimum required ballistic limits - caliber 20 mm fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
1.310	1897	1957	1.470	2205	2320
1.315	1907	1968	1.475	2215	2332
1.320	1917	1978	1.480	2225	2345
1.325	1928	1989	1.485	2235	2357
1.330	1938	2000	1.490	2245	2369
1.335	1948	2010	1.495	2256	2382
1.340	1958	2021	1.500 3/	2266	2395
1.345	1969	2032	1.505	2276	2407
1.350	1979	2043	1.510	2286	2420
1.355	1988	2054	1.515	2296	2433
1.360	1997	2065	1.520	2307	2446
1.365	2006	2075	1.525	2317	2459
1.370	2016	2087	1.530	2327	2471
1.375	2025	2098	1.535	2338	---
1.380	2034	2109	1.540	2348	---
1.385	2043	2120	1.545	2359	---
1.390	2052	2131	1.550	2370	---
1.395	2062	2142	1.555	2380	---
1.400	2071	2154	1.560	2391	---
1.405	2080	2165	1.565	2402	---
1.410	2090	2177	1.570	2412	---
1.415	2099	2189	1.575	2423	---
1.420	2108	2200	1.580	2434	---
1.425	2118	2212	1.585	2445	---
1.430	2128	2224	1.590	2456	---
1.435	2137	2235	1.595	2467	---
1.440	2147	2247	1.600	2478	---
1.445	2156	2259	1.605	2489	---
1.450	2166	2271	1.610	2500	---
1.455	2176	2283	1.615	2511	---
1.460	2186	2295	1.620	2522	---
1.465	2196	2308	1.625	2534	---

3/ Specification type I, class B requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits - caliber 20 mm fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
1.630	2545	---	1.695	2697	---
1.635	2556	---	1.700 4/	2709	---
1.640	2568	---	1.705	2721	---
1.645	2579	---	1.710	2733	---
1.650	2591	---	1.715	2745	---
1.655	2602	---	1.720	2757	---
1.660	2614	---	1.725	2770	---
1.665	2626	---	1.730	2782	---
1.670	2637	---	1.735	2794	---
1.675	2649	---	1.740	2807	---
1.680	2661	---	1.745	2819	---
1.685	2673	---	1.750	2832	---
1.690	2685	---	1.755	---	---

4/ Specification type I, class A requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - caliber 30 APM2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
1.000	1785	2046	1.150	1926	2222
1.005	1789	2052	1.155	1931	2227
1.010	1794	2058	1.160	1936	2233
1.015	1799	2064	1.165	1940	2239
1.020	1804	2070	1.170	1945	2244
1.025	1808	2076	1.175	1950	2250
1.030	1813	2082	1.180	1955	2255
1.031 1/	1814	2083	1.185	1959	2261
1.035	1818	2088	1.190	1964	2266
1.040	1822	2094	1.195	1969	2272
1.045	1827	2100	1.200	1973	2277
1.050	1832	2106	1.205	1978	2283
1.055	1837	2112	1.210	1983	2288
1.060	1841	2118	1.215	1988	2294
1.065	1846	2124	1.220	1992	2299
1.070	1851	2130	1.225	1997	2305
1.075	1856	2136	1.230	2002	2310
1.080	1860	2141	1.235	2006	2316
1.085	1865	2147	1.240	2011	2321
1.090	1870	2153	1.245	2016	2326
1.095	1874	2159	1.250	2021	2332
1.100	1879	2165	1.255	2025	2337
1.101 2/	1880	2166	1.260	2030	2343
1.105	1884	2171	1.265	2035	2348
1.110	1889	2176	1.270	2040	2353
1.115	1893	2182	1.275	2044	2359
1.120	1898	2188	1.280	2049	2364
1.125	1903	2193	1.285	2054	2369
1.130	1907	2199	1.290	2058	2375
1.135	1912	2205	1.295	2063	2380
1.140	1917	2210	1.300	2068	2385
1.145	1922	2216	1.305	2073	2390

1/ Specification type I, class B requirements begin for this ordered thickness.
2/ Specification type I, class A requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - caliber 30 APM2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
1.310	2077	2395	1.470	2231	2558
1.315	2082	2401	1.475	2235	2563
1.320	2087	2406	1.480	2240	2568
1.325	2091	2411	1.485	2245	2573
1.330	2096	2416	1.490	2249	2577
1.335	2101	2422	1.495	2254	2582
1.340	2106	2427	1.500 3/	2259	2587
1.345	2110	2432	1.505	2263	2592
1.350	2115	2437	1.510	2267	2597
1.355	2120	2442	1.515	2272	2602
1.360	2125	2447	1.520	2276	2607
1.365	2129	2452	1.525	2281	2612
1.370	2134	2458	1.530	2286	2616
1.375	2140	2463	1.535	2291	2621
1.380	2145	2468	1.540	2296	2626
1.385	2150	2473	1.545	2300	2631
1.390	2154	2478	1.550	2305	2635
1.395	2159	2483	1.555	2309	---
1.400	2164	2488	1.560	2313	---
1.405	2169	2493	1.565	2318	---
1.410	2174	2498	1.570	2323	---
1.415	2178	2503	1.575	2327	---
1.420	2183	2508	1.580	2332	---
1.425	2188	2513	1.585	2336	---
1.430	2193	2518	1.590	2341	---
1.435	2198	2523	1.595	2345	---
1.440	2202	2528	1.600	2350	---
1.445	2207	2533	1.605	2354	---
1.450	2211	2538	1.610	2359	---
1.455	2216	2543	1.615	2363	---
1.460	2221	2548	1.620	2367	---
1.465	2226	2553	1.625	2371	---

3/ Specification type I, class B requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - caliber 30 APM2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
1.630	2375	---	1.695	2432	---
1.635	2379	---	1.700 4/	2436	---
1.640	2384	---	1.705	2440	---
1.645	2388	---	1.710	2445	---
1.650	2393	---	1.715	2449	---
1.655	2397	---	1.720	2453	---
1.660	2401	---	1.725	2457	---
1.665	2406	---	1.730	2462	---
1.670	2410	---	1.735	2467	---
1.675	2415	---	1.740	2471	---
1.680	2419	---	1.745	2475	---
1.685	2423	---	1.750	2479	---
1.690	2427	---	1.755	---	---

4/ Specification type I, class A requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .50 APM2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
1.450	---	1921	1.610	---	2052
1.455	---	1925	1.615	---	2056
1.460	---	1929	1.620	---	2060
1.465	---	1933	1.625	---	2064
1.470	---	1938	1.630	---	2068
1.475	---	1942	1.635	---	2072
1.480	---	1946	1.640	---	2075
1.485	---	1950	1.645	---	2079
1.490	---	1954	1.650	---	2083
1.495	---	1958	1.655	---	2087
1.500	---	1962	1.660	---	2091
1.501 [1/]	---	1963	1.665	---	2095
1.505	---	1967	1.670	---	2099
1.510	---	1971	1.675	1807	2103
1.515	---	1975	1.680	1811	2107
1.520	---	1979	1.685	1815	2110
1.525	---	1983	1.690	1819	2114
1.530	---	1987	1.695	1823	2118
1.535	---	1991	1.700	1826	2122
1.540	---	1995	1.701 [2/]	1827	2123
1.545	---	2000	1.705	1830	2136
1.550	---	2004	1.710	1834	2130
1.555	---	2008	1.715	1838	2134
1.560	---	2012	1.720	1841	2137
1.565	---	2016	1.725	1845	2141
1.570	---	2020	1.730	1849	2145
1.575	---	2024	1.735	1852	2149
1.580	---	2028	1.740	1856	2153
1.585	---	2032	1.745	1860	2157
1.590	---	2036	1.750	1863	2160
1.595	---	2040	1.755	1867	2164
1.600	---	2044	1.760	1871	2168
1.605	---	2048	1.765	1874	2171

[1/] Specification type I, class B requirements begin for this ordered thickness.
[2/] Specification type I, class A requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .50 APM2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
1.770	1878	2175	1.940	1998	2299
1.775	1882	2179	1.945	2002	2303
1.780	1885	2183	1.950	2005	2306
1.785	1889	2186	1.955	2008	2310
1.790	1893	2190	1.960	2012	2313
1.795	1896	2194	1.965	2015	2317
1.800	1900	2198	1.970	2019	2320
1.805	1903	2201	1.975	2022	2324
1.810	1907	2205	1.980	2025	2328
1.815	1911	2209	1.985	2029	2331
1.820	1914	2212	1.990	2032	2334
1.825	1918	2216	1.995	2036	2338
1.830	1921	2220	2.000	2039	2342
1.835	1925	2223	2.005	2042	2345
1.840	1928	2227	2.010	2046	2349
1.845	1932	2231	2.015	2049	2352
1.850	1936	2234	2.020	2052	2355
1.855	1939	2238	2.025	2056	2359
1.860	1943	2242	2.030	2059	2362
1.865	1946	2245	2.035	2062	2366
1.870	1950	2249	2.040	2066	2369
1.875	1953	2253	2.045	2069	2373
1.880	1957	2256	2.050	2072	2376
1.885	1960	2260	2.055	2076	2380
1.890	1964	2263	2.060	2079	2383
1.895	1967	2267	2.065	2082	2386
1.900	1971	2271	2.070	2085	2390
1.905	1974	2274	2.075	2089	2393
1.910	1978	2278	2.080	2092	2397
1.915	1981	2281	2.085	2095	2400
1.920	1984	2285	2.090	2098	2404
1.925	1988	2289	2.095	2102	2407
1.930	1991	2292	2.100	2105	2410
1.935	1995	2296	2.105	2108	2414

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .50 APM2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
2.110	2111	2417	2.280	2219	2530
2.115	2115	2420	2.285	2222	2533
2.120	2118	2424	2.290	2225	2536
2.125	2121	2427	2.295	2228	2539
2.130	2124	2431	2.300	2231	2542
2.135	2128	2434	2.305	2234	2546
2.140	2131	2437	2.310	2237	2549
2.145	2134	2441	2.315	2240	2552
2.150	2137	2444	2.320	2244	2555
2.155	2140	2447	2.325	2247	2559
2.160	2144	2451	2.330	2250	2562
2.165	2147	2454	2.335	2253	2565
2.170	2150	2457	2.340	2256	2568
2.175	2153	2461	2.345	2259	2571
2.180	2156	2464	2.350	2262	2575
2.185	2160	2467	2.355	2265	2578
2.190	2163	2471	2.360	2268	2581
2.195	2166	2474	2.365	2271	2584
2.200	2169	2477	2.370	2274	2587
2.205	2172	2481	2.375	2277	2590
2.210	2175	2484	2.380	2280	2594
2.215	2179	2487	2.385	2283	2597
2.220	2182	2491	2.390	2286	2600
2.225	2185	2494	2.395	2289	2603
2.230	2188	2497	2.400	2292	2606
2.235	2191	2500	2.405	2295	2609
2.240	2194	2504	2.410	2298	2612
2.245	2197	2507	2.415	2301	2616
2.250	2200	2510	2.420	2304	2619
2.255	2204	2513	2.425	2307	2622
2.260	2207	2517	2.430	2310	2625
2.265	2210	2520	2.435	2313	2628
2.270	2213	2523	2.440	2316	2631
2.275	2216	2526	2.445	2319	2634

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .50 APM2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
2.450	2322	2638	2.615	2417	2738
2.455	2324	2641	2.620	2420	2741
2.460	2327	2644	2.625	2423	2744
2.465	2330	2647	2.630	2425	2747
2.470	2333	2650	2.635	2428	2750
2.475	2336	2653	2.640	2431	2753
2.480	2339	2656	2.645	2434	2756
2.485	2342	2659	2.650	2437	2759
2.490	2345	2662	2.655	2439	2762
2.495	2348	2665	2.660	2442	2765
2.500	2351	2668	2.665	2445	2768
2.505	2354	2672	2.670	2448	2771
2.510	2357	2675	2.675	2451	2774
2.515	2360	2678	2.680	2453	2777
2.520	2362	2681	2.685	2456	2780
2.525	2365	2684	2.690	2459	2783
2.530	2368	2687	2.695	2462	2786
2.535	2371	2690	2.700	2465	2789
2.540	2374	2693	2.705	2467	2792
2.545	2377	2696	2.710	2470	2795
2.550	2380	2699	2.715	2473	2798
2.555	2383	2702	2.720	2476	2800
2.560	2386	2705	2.725	2478	2803
2.565	2388	2708	2.730	2481	2806
2.570	2391	2711	2.735	2484	2809
2.575	2394	2714	2.740	2487	2812
2.580	2397	2717	2.745	2489	2815
2.585	2400	2720	2.750	2492	2818
2.590	2403	2723	2.755	2495	2821
2.595	2406	2726	2.760	2498	2824
2.600	2408	2729	2.765	2500	2827
2.605	2411	2732	2.770	2503	2830
2.610	2414	2735	2.775	2506	2832

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .50 APM2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type I, Class A	Type I, Class B		Type I, Class A	Type I, Class B
2.780	2509	2835	2.945	2597	2930
2.785	2511	2838	2.950	2600	2932
2.790	2514	2841	2.955	2602	2935
2.795	2517	2844	2.960	2605	2938
2.800	2520	2847	2.965	2608	2941
2.805	2522	2850	2.970	2610	2944
2.810	2525	2853	2.975	2613	2946
2.815	2528	2856	2.980	2616	2949
2.820	2530	2858	2.985	2618	2952
2.825	2533	2861	2.990	2621	2954
2.830	2536	2864	2.995	2623	2957
2.835	2538	2867	3.000 3/	2626	2960
2.840	2541	2870	3.005	2629	2963
2.845	2544	2873	3.010	2631	2966
2.850	2547	2876	3.015	2634	2969
2.855	2549	2879	3.020	2636	2972
2.860	2552	2881	3.025	2639	2975
2.865	2555	2884	3.030	2642	2978
2.870	2557	2887	3.035	2644	---
2.875	2560	2890	3.040	2647	---
2.880	2563	2893	3.045	2649	---
2.885	2565	2896	3.050	2652	---
2.890	2568	2898	3.055	2654	---
2.895	2571	2901	3.060	2657	---
2.900	2573	2904	3.065	2660	---
2.905	2576	2907	3.070	2662	---
2.910	2579	2910	3.075	2665	---
2.915	2581	2913	3.080	2667	---
2.920	2584	2915	3.085	2670	---
2.925	2587	2918	3.090	2672	---
2.930	2589	2921	3.095	2675	---
2.935	2592	2924	3.100	2678	---
2.940	2594	2927	3.105	---	---

3/ Specification type I, class A and class B requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
2.900	TBD	3.065	TBD	3.235	TBD
2.905	TBD	3.070	TBD	3.240	TBD
2.910	TBD	3.075	TBD	3.245	TBD
2.915	TBD	3.080	TBD	3.250	TBD
2.920	TBD	3.085	TBD	3.255	TBD
2.925	TBD	3.090	TBD	3.260	TBD
2.930	TBD	3.095	TBD	3.265	TBD
2.935	TBD	3.100	TBD	3.270	TBD
2.940	TBD	3.105	TBD	3.275	TBD
2.945	TBD	3.110	TBD	3.280	TBD
2.950	TBD	3.115	TBD	3.285	TBD
2.955	TBD	3.120	TBD	3.290	TBD
2.960	TBD	3.125	TBD	3.295	TBD
2.965	TBD	3.130	TBD	3.300	TBD
2.970	TBD	3.135	TBD	3.305	TBD
2.975	TBD	3.140	TBD	3.310	TBD
2.980	TBD	3.145	TBD	3.315	TBD
2.985	TBD	3.150	TBD	3.320	TBD
2.990	TBD	3.155	TBD	3.325	TBD
2.995	TBD	3.160	TBD	3.330	TBD
3.000	TBD	3.165	TBD	3.335	TBD
3.001 1/	TBD	3.170	TBD	3.340	TBD
3.005	TBD	3.175	TBD	3.345	TBD
3.010	TBD	3.180	TBD	3.350	TBD
3.015	TBD	3.185	TBD	3.355	TBD
3.020	TBD	3.190	TBD	3.360	TBD
3.025	TBD	3.195	TBD	3.365	TBD
3.030	TBD	3.200	TBD	3.370	TBD
3.035	TBD	3.205	TBD	3.375	TBD
3.040	TBD	3.210	TBD	3.380	TBD
3.045	TBD	3.215	TBD	3.385	TBD
3.050	TBD	3.220	TBD	3.390	TBD
3.055	TBD	3.225	TBD	3.395	TBD
3.060	TBD	3.230	TBD	3.400	TBD

1/ Specification type I, class B requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
3.405	TBD	3.570	TBD	3.715	TBD
3.410	TBD	3.575	TBD	3.720	TBD
3.415	TBD	3.580	TBD	3.725	TBD
3.420	TBD	3.585	TBD	3.730	TBD
3.425	TBD	3.590	TBD	3.735	TBD
3.430	TBD	3.595	TBD	3.740	TBD
3.435	TBD	3.600	TBD	3.745	TBD
3.440	TBD	3.605	TBD	3.750	TBD
3.445	TBD	3.610	TBD	3.755	TBD
3.450	TBD	3.615	TBD	3.760	TBD
3.455	TBD	3.620	TBD	3.765	TBD
3.460	TBD	3.625	TBD	3.770	TBD
3.465	TBD	3.630	TBD	3.775	TBD
3.470	TBD	3.635	TBD	3.780	TBD
3.475	TBD	3.640	TBD	3.785	TBD
3.480	TBD	3.645	TBD	3.790	TBD
3.485	TBD	3.650	TBD	3.795	TBD
3.490	TBD	3.655	TBD	3.800	TBD
3.495	TBD	3.660	TBD	3.805	TBD
3.500	TBD	3.665	TBD	3.810	TBD
3.505	TBD	3.670	TBD	3.815	TBD
3.510	TBD	3.675	TBD	3.820	TBD
3.515	TBD	3.680	TBD	3.825	TBD
3.520	TBD	3.685	TBD	3.830	TBD
3.525	TBD	3.670	TBD	3.835	TBD
3.530	TBD	3.675	TBD	3.840	TBD
3.535	TBD	3.680	TBD	3.845	TBD
3.540	TBD	3.685	TBD	3.850	TBD
3.545	TBD	3.690	TBD	3.855	TBD
3.550	TBD	3.695	TBD	3.860	TBD
3.555	TBD	3.700	TBD	3.865	TBD
3.560	TBD	3.705	TBD	3.870	TBD
3.565	TBD	3.710	TBD	3.875	TBD

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - 14.5 mm BS41 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
3.880	TBD	3.940	TBD	4.000 2/	TBD
3.885	TBD	3.945	TBD	4.005	TBD
3.890	TBD	3.950	TBD	4.010	TBD
3.895	TBD	3.955	TBD	4.015	TBD
3.900	TBD	3.960	TBD	4.020	TBD
3.905	TBD	3.965	TBD	4.025	TBD
3.910	TBD	3.970	TBD	4.030	TBD
3.915	TBD	3.975	TBD	4.035	TBD
3.920	TBD	3.980	TBD	4.040	TBD
3.925	TBD	3.985	TBD	4.045	TBD
3.930	TBD	3.990	TBD	4.050	TBD
3.935	TBD	3.995	TBD	4.055	---

2/ Specification type I, class B requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
ARMY - MR

(Project FORG-2025-001)

Review activities:
Army - AR, AT, AV, TE

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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